Worksheet 4.1 — Introducing Quadratic Relations

Grade 10 Academic (MPM2D) · Unit 4: Quadratic Relations

The Big Ideas

A **quadratic relation** has degree 2 and graphs as a **parabola**. In a table with equal x-steps, its **second differences are constant**. Use the difference table to test a relation, find missing values, find a, b and c — and always **check the points on the graph**.

- **1st differences** = differences of neighbouring y-values; **2nd differences** = differences of those.
- Constant 1st differences ⇒ linear. Constant (non-zero) 2nd differences ⇒ **quadratic**.
- With x-steps of size h: 2nd difference $= 2ah^2$ (so $2a$ when $h = 1$). Then $c = y(0)$ and one more point gives b.
- A point (x, y) is on the graph exactly when substituting x gives y. A table row off the curve means a wrong value.

Concept Spotlight

Difference tables

Subtract neighbouring y-values to get the **1st differences**, and subtract again for the **2nd differences**. Linear relations have constant 1st differences; **quadratic** relations have constant 2nd differences.

From a table to the equation

With equal x-steps of size h, the 2nd difference is $2ah^2$ (just $2a$ when $h = 1$). Then $c = y(0)$, and one more point gives b. Check with a point you did not use.

Check the point on the graph

A point is on the graph exactly when substituting its x-value gives its y-value. Every table row is a point on the graph, so a row that sits off the curve is a wrong value.

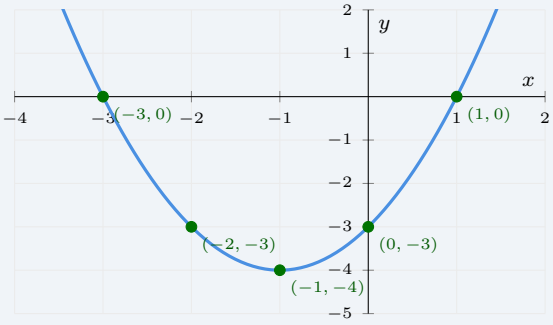
Worked Examples

Example 1 — Build the table and check the graph

Complete the difference table for $y = x^2 + 2x - 3$ and decide whether it is quadratic.

x	-3	-2	-1	0	1
y	0	-3	-4	-3	0
1st diff		-3	-1	$+1$	$+3$
2nd diff			$+2$	$+2$	$+2$

Solution. 1st differences -3, -1, 1, 3; 2nd differences 2, 2, 2 — constant, so quadratic with $2a = 2$, $a = 1$. Check $(1, 0)$: $1 + 2 - 3 = 0$ ✓.



Example 2 — Linear, quadratic or neither?

Use differences to classify each table.

x	0	1	2	3	4
A	-2	1	4	7	10
1st diff		$+3$	$+3$	$+3$	$+3$
2nd diff			$+0$	$+0$	$+0$

x	0	1	2	3	4
B	0	-1	0	3	8
1st diff		-1	$+1$	$+3$	$+5$
2nd diff			$+2$	$+2$	$+2$

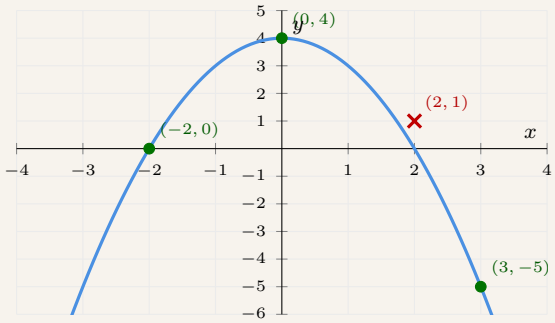
x	0	1	2	3	4
C	1	3	9	27	81
1st diff		$+2$	$+6$	$+18$	$+54$
2nd diff			$+4$	$+12$	$+36$

Solution. **A:** 1st differences all 3 — linear. **B:** 1st differences $-1, 1, 3, 5$ change, 2nd differences all 2 — quadratic. **C:** 1st differences $2, 6, 18, 54$ and 2nd differences $4, 12, 36$ both keep changing — neither.

Example 3 — Is the point on the graph?

The graph is $y = -x^2 + 4$. Is $(3, -5)$ on it? Is $(2, 1)$?

Solution. For $x = 3$: $y = -9 + 4 = -5$, so $(3, -5)$ **is** on the graph. For $x = 2$: $y = -4 + 4 = 0 \neq 1$, so $(2, 1)$ is **not** (the graph passes through $(2, 0)$).



Example 4 — Find the missing value

The relation is quadratic. Find the missing y-value.

x	0	1	2	3	4
y	1	2		16	29
1st diff		+1	+5	+9	+13
2nd diff			+4	+4	+4

Solution. Known 1st differences: 1 (first step) and 13 (last step). If the 2nd difference is d, the last 1st difference is $1 + 3d = 13$, so $d = 4$. The 1st differences are $1, 5, 9, 13$, so the missing value is $2 + 5 = 7$. Check: $7 + 9 = 16$ ✓.

Example 5 — Find a, b and c from a table

Find the equation $y = ax^2 + bx + c$ from the table.

x	0	1	2	3
y	5	6	11	20
1st diff		+1	+5	+9
2nd diff			+4	+4

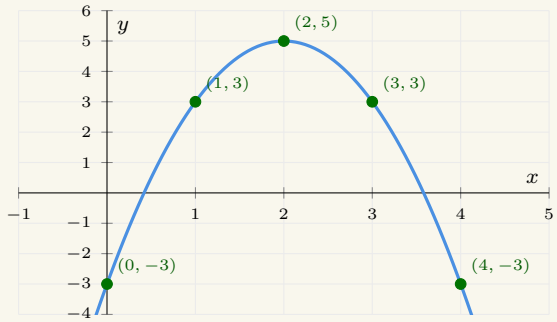
Solution. 2nd difference $4 = 2a$, so $a = 2$. $c = y(0) = 5$. At $x = 1$: $2 + b + 5 = 6$, so $b = -1$. $y = 2x^2 - x + 5$. Check $x = 3$: $18 - 3 + 5 = 20$ ✓.

Example 6 — A parabola that opens down

Use the table to find a, the direction of opening, and the maximum value.

x	0	1	2	3	4
y	-3	3	5	3	-3
1st diff		$+6$	$+2$	-2	-6
2nd diff			-4	-4	-4

Solution. 2nd difference -4, so $a = -2 < 0$: opens **down**. The values rise to 5 at $x = 2$, then fall: maximum value 5, vertex $(2, 5)$.



Example 7 — Steps of 3

The x-values go up by 3. Find a if $y = ax^2$.

x	0	3	6	9
y	0	18	72	162
1st diff		$+18$	$+54$	$+90$
2nd diff			$+36$	$+36$

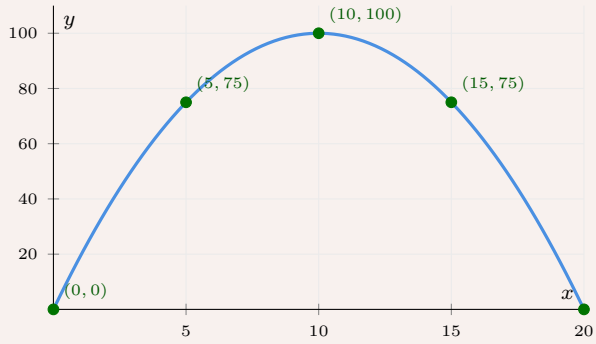
Solution. 1st differences $18, 54, 90$; 2nd differences $36, 36$. With $h = 3$: $2ah^2 = 18a = 36$, so $a = 2$. Check: $y(9) = 2(81) = 162$ ✓.

Example 8 — Maximum area

A fence of 40 m encloses a rectangle of width w, so $A = w(20 - w)$. Study the table.

x	0	5	10	15	20
A	0	75	100	75	0
1st diff		+75	+25	−25	−75
2nd diff			−50	−50	−50

Solution. 2nd differences -50; with $h = 5$: $2ah^2 = 50a = -50$, so $a = -1$ (opens down). The largest area is 100 m^2 at $w = 10$ m (a square 10 by 10).

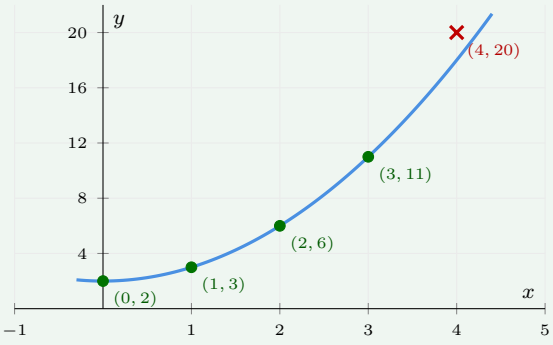


Example 9 — Find the error

A student says this table is quadratic. Find the mistake and the correct value.

x	0	1	2	3	4
y	2	3	6	11	20
1st diff		+1	+3	+5	+9
2nd diff			+2	+2	+4

Solution. 1st differences $1, 3, 5, 9$; 2nd differences $2, 2, 4$ — the last one breaks the pattern. The last 1st difference should be 7, so the last value should be $11 + 7 = 18$, not 20. Check on $y = x^2 + 2$: $y(4) = 18$.



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Complete the differences for each table and decide whether it is linear, quadratic or neither.

x	1	2	3	4	5
A	2	5	10	17	26
1st diff					
2nd diff					

x	0	1	2	3	4
B	-3	1	5	9	13
1st diff					
2nd diff					

Question 2

Complete the table for $y = 2x^2 - 3$, then fill in both difference rows.

x	-2	-1	0	1	2
y					
1st diff					
2nd diff					

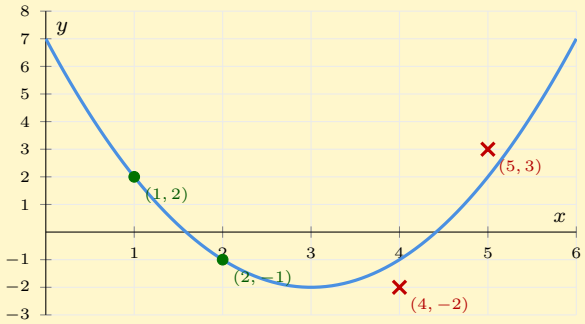
Question 3

This relation is quadratic. Find the missing value.

x	0	1	2	3	4	5
y	4	2	2		8	14
1st diff		-2	$+0$	$+2$	$+4$	$+6$
2nd diff			$+2$	$+2$	$+2$	$+2$

Question 4

The graph of $y = x^2 - 6x + 7$ is shown. Which of $A(1, 2)$, $B(2, -1)$, $C(4, -2)$, $D(5, 3)$ lie on the graph? Correct the y-value of any point that does not.



Question 5

Find the equation $y = ax^2 + bx + c$ of the relation.

x	0	1	2	3
y	1	2	9	22
1st diff		+1	+7	+13
2nd diff			+6	+6

Question 6

Use the table to find a, say whether the parabola opens up or down, and give the vertex.

x	0	1	2	3	4	5	6
y	-4	1	4	5	4	1	-4
1st diff		$+5$	$+3$	$+1$	-1	-3	-5
2nd diff			-2	-2	-2	-2	-2

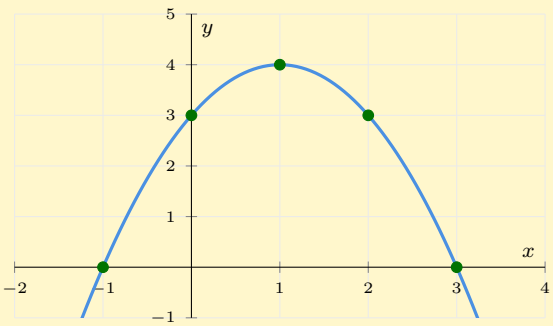
Question 7

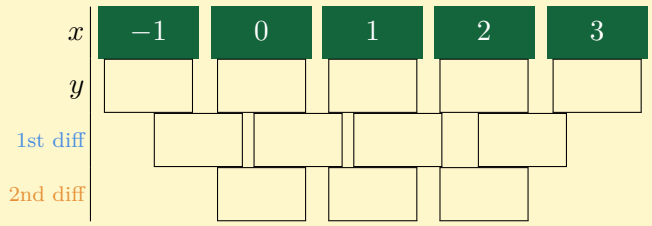
The x-values go up by 2. Find the equation $y = ax^2 + bx + c$.

x	0	2	4	6
y	1	5	17	37
1st diff		$+4$	$+12$	$+20$
2nd diff			$+8$	$+8$

Question 8

Read the points from the graph of $y = -x^2 + 2x + 3$, complete the table and its differences, and state a.



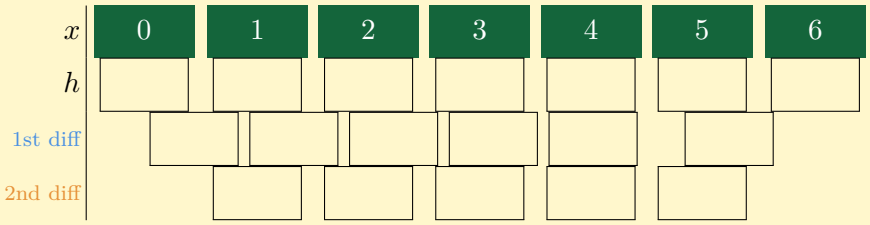


Question 9

For $f(x) = x^2 - 5x + 4$, find $f(-1)$ and $f(6)$. Is $(3, -2)$ on the graph of f? Show your substitution.

Question 10

A ball's height is $h = -5t^2 + 30t$ (metres). Complete the table, find a from the differences, the maximum height, and the time the ball is in the air.



Question 11

One value in this table is wrong. Find it and correct it.

x	0	1	2	3	4
y	1	4	9	17	25
1st diff		+3	+5	+8	+8
2nd diff			+2	+3	+0

Question 12

Show that for $y = ax^2 + bx + c$ and $x = 0, 1, 2$, the second difference is $2a$.

Question 13 — Challenge

A quadratic relation passes through $(0, -5)$, $(1, -2)$ and $(2, 5)$. Find y when $x = 10$, and say whether $(3, 16)$ is on the graph.

Question Recap & Answer Key

Question Recap

1. Complete the differences for each table and decide whether it is linear, quadratic or neither.

x	1	2	3	4	5
A	2	5	10	17	26
1st diff					
2nd diff					

x	0	1	2	3	4
B	−3	1	5	9	13
1st diff					
2nd diff					

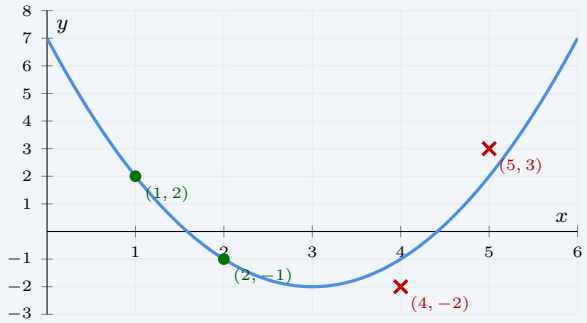
2. Complete the table for $y = 2x^2 - 3$, then fill in both difference rows.

x	−2	−1	0	1	2
y					
1st diff					
2nd diff					

3. This relation is quadratic. Find the missing value.

x	0	1	2	3	4	5
y	4	2	2		8	14
1st diff	−2	+0	+2	+4	+6	
2nd diff	+2	+2	+2	+2		

4. The graph of $y = x^2 - 6x + 7$ is shown. Which of $A(1, 2)$, $B(2, -1)$, $C(4, -2)$, $D(5, 3)$ lie on the graph? Correct the y-value of any point that does not.



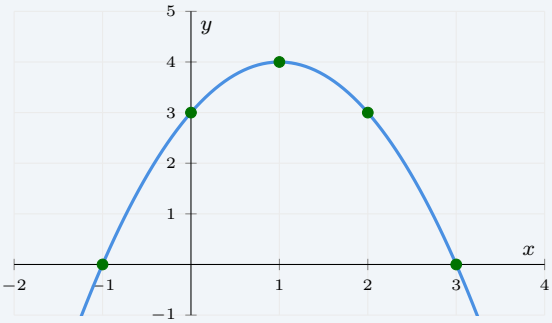
5. Find the equation $y = ax^2 + bx + c$ of the relation.

x	0	1	2	3
y	1	2	9	22
1st diff		+1	+7	+13
2nd diff			+6	+6

7. The x-values go up by 2. Find the equation $y = ax^2 + bx + c$.

x	0	2	4	6
y	1	5	17	37
1st diff		+4	+12	+20
2nd diff			+8	+8

8. Read the points from the graph of $y = -x^2 + 2x + 3$, complete the table and its differences, and state a.



x	−1	0	1	2	3
y					
1st diff					
2nd diff					

9. For $f(x) = x^2 - 5x + 4$, find $f(-1)$ and $f(6)$. Is $(3, -2)$ on the graph of f? Show your substitution.

10. A ball's height is $h = -5t^2 + 30t$ (metres). Complete the table, find a from the differences, the maximum height, and the time the ball is in the air.

x	0	1	2	3	4	5
h						
1st diff						
2nd diff						

11. One value in this table is wrong. Find it and correct it.

x	0	1	2	3	4
y	1	4	9	17	25
1st diff		+3	+5	+8	+8

✓ Answer Key

1. A: 1st $3, 5, 7, 9$; 2nd $2, 2, 2$ — quadratic. B: 1st $4, 4, 4, 4$ — linear.
2. y: $5, -1, -3, -1, 5$; 1st: $-6, -2, 2, 6$; 2nd: $4, 4, 4$ (so $a = 2$).
3. y-values $4, 2, 2, ?, 8, 14$; 1st differences $-2, 0, 2, 4, 6$: the missing value is $2 + 2 = 4$.
4. A and B are on the graph. $y(4) = -1$, so C should be $(4, -1)$; $y(5) = 2$, so D should be $(5, 2)$.
5. 2nd difference 6, so $a = 3$; $c = 1$; at $x = 1$: $3 + b + 1 = 2$, so $b = -2$. $y = 3x^2 - 2x + 1$.
6. 2nd difference -2, so $a = -1$ (opens down). Vertex $(3, 5)$.
7. 2nd difference $8 = 2a(2)^2$, so $a = 1$; $c = 1$; at $x = 2$: $4 + 2b + 1 = 5$, so $b = 0$. $y = x^2 + 1$.
8. Table y: $0, 3, 4, 3, 0$; 1st: $3, 1, -1, -3$; 2nd: $-2, -2, -2$; $a = -1$.
9. $f(-1) = 1 + 5 + 4 = 10$; $f(6) = 36 - 30 + 4 = 10$; $f(3) = 9 - 15 + 4 = -2$, so $(3, -2)$ **is** on the graph.
10. h: $0, 25, 40, 45, 40, 25, 0$; 2nd difference -10, so $a = -5$; maximum 45 m at $t = 3$ s; in the air 6 s.
11. 2nd differences $2, 3, 0$ are not constant. The value 17 should be 16 (then 1st differences $3, 5, 7, 9$).
12. $y(0) = c$, $y(1) = a + b + c$, $y(2) = 4a + 2b + c$. 1st: $a + b$ and $3a + b$. 2nd: $(3a + b) - (a + b) = 2a$.
13. 2nd difference 4, so $a = 2$; $c = -5$; $2 + b - 5 = -2$ gives $b = 1$. $y = 2x^2 + x - 5$; $y(10) = 205$. At $x = 3$: $18 + 3 - 5 = 16$, so $(3, 16)$ is on the graph.